

SQL Performance Tips

CHEAT SHEET — Performance Optimization

FETCHING DATA

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FETCHING DATA

Tips #1 – #3

- **#1 Only select the columns you need.**
Avoid using SELECT *
- **#2 Avoid DISTINCT or ORDER BY unless necessary,**
they can slow down queries
- **#3 For exploration, limit rows using TOP**
to avoid fetching unnecessary data

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FILTERING DATA

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FILTERING DATA

Tips #4 – #7

- #4 **Create non-clustered indexes on columns in WHERE**
to speed up queries
- #5 **Avoid functions like UPPER(), YEAR() in WHERE —**
they prevent indexes from being used
- #6 **Avoid starting string searches with a wildcard (%example),**
this disables index usage
- #7 **Use IN instead of multiple OR conditions**
for better readability and performance

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JOINING DATA

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JOINING DATA

Tips #8 – #16

- **#8 Know the cost of different join types.**
Use INNER JOIN when possible for efficiency
- **#9 Always use explicit ANSI-style joins**
(INNER JOIN, LEFT JOIN) — avoid old implicit syntax
- **#10 Ensure columns in your ON clause are indexed**
for optimal join performance
- **#11 Filter before joining large tables**
to reduce the size of the dataset being joined
- **#12 Aggregate before joining large tables**
to reduce the size of the dataset being joined
- **#13 Replace OR conditions in join logic with UNION**
to improve query performance
- **#14 Watch for nested loops in your execution plan.**
Use SQL Hints if needed
- **#15 Use UNION ALL instead of UNION when duplicates are OK —**
it's faster
- **#16 When dups aren't OK, use UNION ALL + DISTINCT**
instead of UNION for better performance

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AGGREGATING DATA

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AGGREGATING DATA

Tips #17 – #18

- #17 **Use columnstore indexes for queries with heavy aggregations on large tables**
- #18 **Pre-aggregate data and store results in a separate table for faster reporting**

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SUBQUERIES

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SUBQUERIES

Tips #19 – #20



#19

Know when to use JOIN, EXISTS, or IN.

Avoid IN with large lists — it can be inefficient



#20

Simplify queries by eliminating redundant logic using CTEs

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DDL

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DDL

Tips #21 – #25

- #21 **Avoid VARCHAR or TEXT types unnecessarily —**
use precise data types to save storage and boost performance
- #22 **Avoid excessive lengths like VARCHAR(MAX)**
unless truly needed
- #23 **Use NOT NULL constraints wherever possible**
to enforce data integrity
- #24 **Ensure all tables have a clustered primary key**
for structure and query performance
- #25 **Add non-clustered indexes to foreign keys**
that are frequently queried

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INDEXING

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INDEXING

Tips #26 – #30

- #26 **Avoid over-indexing —**
it slows down INSERT, UPDATE, and DELETE operations
- #27 **Regularly review and drop unused indexes**
to save space and improve write performance
- #28 **Update table statistics weekly**
so the query optimizer has current information
- #29 **Reorganize and rebuild fragmented indexes weekly**
to maintain query performance
- #30 **For large tables, partition the data**
then apply a columnstore index for best results